

Emerging Technologies -- Guided Notes ANSWER KEY

Unit tp-4.7 | CompTIA Network+ N10-009 Obj. FC0-U71 4.7 | N10-008 FC0-U71 4.7

For instructor use / distribute after students complete the notes.

Question 1

Augmented Reality (AR) _____ digital content onto the real world (visible through a phone camera or AR glasses). Virtual Reality (VR) creates a fully _____ digital environment that replaces the physical world. Mixed Reality (MR) allows digital objects to interact with _____ world objects.

Answer: Overlays; immersive; real.

Question 2

Blockchain is a _____-distributed ledger that records transactions across multiple nodes. Each block contains a _____ of the previous block, creating a chain that makes tampering _____ to detect. It is the underlying technology for _____ currencies like Bitcoin and Ethereum.

Answer: Decentralized; cryptographic hash; easy; cryptocurrency.

Question 3

Quantum computing uses _____ (quantum bits) that can exist as 0, 1, or both simultaneously (superposition). This allows quantum computers to evaluate _____ exponentially more solutions at once than classical computers. A sufficiently powerful quantum computer could break current _____ encryption (RSA/ECC), requiring _____-resistant algorithms.

Answer: Qubits; many; public-key; quantum (post-quantum).

Question 4

5G is the 5th generation cellular standard. Compared to 4G LTE, 5G offers _____ speeds, _____ latency, and supports far more _____ devices simultaneously. Use cases include smart city infrastructure, autonomous vehicles, and _____ (remote robotic surgery). 5G mmWave offers highest speeds but limited _____ penetration.

Answer: Higher; lower; connected; telemedicine; building (indoor).

Question 5

Edge computing moves processing closer to where data is _____, reducing _____ and bandwidth usage. It complements rather than replaces cloud computing. Industrial IoT sensors, self-driving cars, and smart factories use edge processing because they cannot tolerate the _____ ms round-trip delay to a remote cloud server for real-time decisions.

Answer: Generated; latency; 100+.

Question 6

3D printing (additive manufacturing) builds _____ objects layer by layer from a digital _____ (STL file). It enables rapid _____ (quick physical models of designs), customized medical implants, and on-demand part replacement. The two main materials are _____ (plastic filament, FDM) and

_____ (photopolymer, resin).

Answer: Physical; model/file; prototyping; FDM plastic; resin (SLA).

Question 7

Automation replaces repetitive human tasks with machines or software. _____ Process Automation (RPA) uses software robots to automate rule-based digital tasks like data entry. _____ automation uses physical robots for repetitive factory tasks. The societal concern is _____ displacement -- jobs replaced without equivalent new jobs being created.

Answer: Robotic; industrial; workforce.

Question 8 -- Real World Application

REAL WORLD: A small manufacturing company is considering adopting robotics to automate their assembly line. Currently 12 workers perform repetitive welding and packaging tasks. A manager argues the robots will pay for themselves in 3 years and improve product consistency. A workers representative argues it will harm the employees. Describe two legitimate benefits and two legitimate concerns, then suggest one way the company could balance both interests.

Answer: Benefits: (1) Productivity and consistency -- robots work 24/7 without fatigue, producing more consistent output quality and higher throughput. (2) Return on investment -- after the break-even point, labor cost savings are significant and ongoing; production quality improvements reduce defect waste. Concerns: (1) Job displacement -- 12 employees lose income; in regions with few alternative jobs this causes real hardship for families. (2) Concentration of gains -- productivity gains go to owners; workers bear the cost. Balancing approach: commit to retraining employees for roles that cannot be automated (robot maintenance, quality control, programming) and offer transition support. Some companies fund retraining programs and give displaced workers first priority for new technical roles created by the automation. This is not perfect but distributes the benefit more equitably.